

Question				Marking details	Marks available																																
					AO1	AO2	AO3	Total	Maths	Prac																											
3	(a)			<table><tr><th>Property of the ball</th><th>Stays constant</th><th>Gets smaller</th><th>Gets larger</th></tr><tr><td>acceleration</td><td>✓</td><td></td><td></td></tr><tr><td>speed</td><td></td><td>✓</td><td></td></tr><tr><td>kinetic energy</td><td></td><td>✓</td><td></td></tr><tr><td>potential energy</td><td></td><td></td><td>✓</td></tr><tr><td>total energy</td><td>✓</td><td></td><td></td></tr></table>				Property of the ball	Stays constant	Gets smaller	Gets larger	acceleration	✓			speed		✓		kinetic energy		✓		potential energy			✓	total energy	✓			4			4		
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				1 mark for each correct row																																	
	(b)	(i)		Substitution: 0.46×10 (1) = 4.6 [N] (1)				1	1		2	2																									
		(ii)		Substitution: 4.6 (ecf) $\times 15$ (1) = 69 (1) Joules or J or N m (1)				1 1	1		3	2																									
		(iii)		[Yes because] acceleration [due to gravity] is the same as gravitational field strength or no other forces acting or $a = g$ or weight = 0.46×10 also or weight is 4.6 [N] or gravitational field strength = 10 or g is 10 Accept they're both 10 or we only multiplied by 10						1	1																										
				Question 3 total				7	2	1	10	4	0																								